

Using R to Mine *and* Analyse Popular Sentiments

Public opinion is important to enterprises, politicians, governments, film stars and most of us. Opinions can be found on various social media sites. This article demonstrates how popular sentiments can be mined from Twitter with the help of the R programming language.



Opinion mining is an area of text mining where data analytics extracts opinions, emotions and sentiments from a corpus of text. That is why it is also known as sentiment analysis. One of the most common applications of opinion mining is to track the attitudes and moods on the Web, especially to survey the status of products, services, brands or even the consumers. The main purpose is to know whether what is being surveyed is viewed positively or negatively by a given audience.

The R language supports the entire text mining paradigm with its various packages. We will discuss only opinion mining on Twitter data here, and only through packages which concern this topic.

To begin with, let's consider the Sentiment package of Timothy Jurka. The package *sentiment_0.1.tar.gz* is available on <https://cran.r-project.org/src/contrib/Archive/sentiment/>. After downloading it, install the package with *load_all()*. The Sentiment package supports different tools to analyse the text corpus, and assigns weightage to each sentence

on the basis of the sentiment values of different emotional keywords. Since micro bloggers use Twitter to express their thoughts in a casual way, it is a good place for sentiment analysis on different current topics. This information is gathered on the basis of different emotional keywords within the Twitter text. There are two functions that analyse the sentiment measure of a sentence.

1. The function *classify_emotion()* helps us to analyse some text and classify it under different types of emotions. The predefined emotions are anger, disgust, fear, joy, sadness, and surprise. Classifying text into these categories is done using one of the following two algorithms:
 - i. The Naive Bayes classifier trained on Carlo Strapparava and Alessandro Valitutti's emotions lexicon
 - ii. The voter procedure
2. The *classify_polarity()* function performs the classification on the basis of some symbolic enumeration values. The symbolic values are defined as very positive,